Perimeter & Area

Year 4 · **Homework Worksheet**

Worked Example

Worked example: A rectangle is 7 cm long and 3 cm wide. What is its perimeter and area?

Answer: Perimeter = 2×(7+3) = 20 cm. Area = 7×3 = 21 cm².

Easy / Standard (15 questions)

1. A rectangle is 6 cm by 3 cm. What is its perimeter?
2. A square has sides of 5 cm. What is its perimeter?
3. A rectangle is 7 cm by 2 cm. What is its area?
4. Which shape has a perimeter of 16cm? A) a 5cm×3cm rectangle B) a square of side 4cm C) both A and B D) neither
5. A rectangle is 9 cm long and 3 cm wide. What is its perimeter?
6. A square has a perimeter of 16 cm. What is the length of one side?
7. What is the area of a 5cm by 5cm square? A) 10cm² B) 20cm² C) 25cm² D) 30cm²
8. A square has an area of 16 cm². What is the length of one side?
9. Select ALL of these shapes that have a perimeter LESS than 20cm: A) a 5cm×3cm rectangle B) a square of side 6cm C) a 4cm×2cm rectangle D) a 7cm×6cm rectangle E) a square of side 5cm
10. A rectangle has a length of 8 cm and a width of 2 cm. What is its perimeter?
11. A rectangle is 10 cm long and 1 cm wide. What is its area?
12. Which rectangle has the larger area? A) 6cm×2cm B) 4cm×4cm C) they are equal D) cannot tell
13. A square has sides of 7 cm. What is its perimeter?
14. A rectangle is 6 cm by 6 cm. What shape is it, and what is its area?
15. A rectangle has an area of 15 cm² and a width of 3 cm. What is its length?

Standard (15 questions)

1. A rectangle is 8 cm by 5 cm. What is its perimeter?
2. Which rectangle has a perimeter of 24cm? A) 6cm×6cm B) 9cm×3cm C) both A and B D) neither
3. A rectangle is 7 cm by 6 cm. What is its area?
4. A square has an area of 36 cm². What is its perimeter?
5. A rectangle has a perimeter of 30 cm and a length of 9 cm. What is its width?
6. What is the area of a rectangle 8cm long and 5cm wide? A) 13cm² B) 30cm² C) 40cm² D) 45cm²
7. A square garden has sides of 15 m. What is its perimeter?
8. Select ALL of these shapes that have an area of MORE than 24cm²: A) 6cm×5cm B) 4cm×6cm C) 7cm×4cm D) 5cm×5cm E) 3cm×8cm
9. A rectangle is 9 cm long and 5 cm wide. What is its area?

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10. Which shape has the SAME area as a 5cm×6cm rectangle? A) a 3cm×10cm rectangle B) a 4cm×8cm rectangle C) both A and B D) neither

11. A rectangle has a width of 4 cm and its length is triple the width. What is its perimeter?

12. A square has a perimeter of 28 cm. What is its area?

13. Two rectangles, 5cm×2cm and 3cm×2cm, are joined edge to edge with no overlap along their 2cm sides. What is the total area?

14. A rectangular field is 25 m by 12 m. What is its perimeter?

15. A rectangle has an area of 36 cm^2 and a length of 9 cm. What is its width?

Standard / Advanced (15 questions)

1. A rectangle's width is 4 cm. Its length is double the width. What is its perimeter?

2. Compared with a smaller 9cm^2 square, a bigger square covers 4 times as much area. How long is one side of the bigger square? A) 6cm B) 9cm C) 18cm D) 36cm

3. Two identical squares, each with sides of 5cm, are placed side by side to form a rectangle. What is the perimeter of the rectangle?

4. A square has the same perimeter as a 9cm×3cm rectangle. What is the side length of the square?

5. A rectangular lawn is 9 m by 4 m. A 1m×1m flower bed is cut from one corner. What is the remaining area?

6. Two square tiles, each with side 6 cm, are placed side by side touching each other. What is the perimeter of the resulting rectangle?

7. A rectangular room measures 4 m by 3 m. Carpet costs £7 per m^2. What is the total cost of carpeting the room?

8. A square garden has an area of 64 m^2. What is the perimeter of the garden?

9. A large rectangle 8m×5m has a 2m×2m rectangle removed from a corner, forming an L-shape. What is the area of the L-shape? A) 36m^2 B) 38m^2 C) 40m^2 D) 34m^2

10. Select ALL of these statements that are TRUE about a 10cm×4cm rectangle: A) its perimeter is 28cm B) its area is 40cm^2 C) its perimeter is 40cm D) it has 4 equal sides E) its area is 28cm^2

11. A 15cm×10cm piece of card has a 3cm×3cm square cut from one corner. What is the remaining area?

12. A rectangular pitch is 40 m long and 20 m wide. What is the perimeter of the pitch? A) 100m B) 120m C) 140m D) 160m

13. A square and a rectangle have equal areas of 64 cm^2. The rectangle's width is 8 cm. What is the rectangle's length?

14. A room is made of two rectangles, a 4m×3m part and a 2m×2m part, with no overlap. What is the total floor area?

15. A rectangular pool measures 12m by 6m, so its area is 72m^2. It sits inside a 14m by 8m decked area. What is the area of the decking around the pool (not including the pool itself)?

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